FastSpeech 2 Dual-Language Text-to-Speech: English and Telugu

1. Implementation Approach & Architecture

Base Architecture: The project utilizes the FastSpeech 2 feed-forward transformer architecture. It employs a Variance Adaptor consisting of Duration, Pitch (F0), and Energy predictors to generate highly controllable mel-spectrograms from phoneme sequences.**Modifications:**

- **Vocoder:** We replaced the default HiFi-GAN with the **Griffin-Lim** algorithm (model: 'Griffin-Lim' in model.yaml). This eliminated the need for external neural vocoder weights while providing rapid, deterministic phase reconstruction.
- **Checkpoint Handling:** Modified utils/model.py to enforce map_location='cpu' and weights_only=False. This resolved PyTorch 2.6 security unpickling blocks and allowed GPU-trained models to be evaluated seamlessly on CPU environments.
- **Inference Pipeline:** Altered the synthesis inputs to accept raw ARPAbet/Indic phoneme representations enclosed in brackets (e.g., {DH AH0 ...}) directly in single-sentence mode, circumventing text-cleaner misalignment for edge-case words (e.g., "10:00 AM").

2. Dataset Selection & Training Configuration

- **English:** Trained on the **Svarah** dataset.
- **Telugu:** Trained on the **IndicTTS** Telugu dataset.
- **Preprocessing:** Audio was resampled to 22050 Hz. Transcripts were converted to phonemes. Mel-spectrograms, pitch contours (extracted via PyWorld), and energy (L2 norm of amplitude) were computed per frame. Phoneme durations were extracted using Montreal Forced Aligner (MFA) TextGrids.
- **Splits:** A standard 95% Train / 5% Validation split was utilized.
- **Training Configuration:** Both models were trained up to **12,000 steps** with a batch size optimized for GPU memory, utilizing the ScheduledOptim Adam optimizer.

3. Analysis of Loss Curves

- **Convergence Behavior:** Both models exhibited a rapid drop in total loss during the first 2,000 steps, driven primarily by the mel-spectrogram reconstruction loss.
- **Variance Predictor Losses:** Duration loss converged the fastest (around 4,000 steps). Pitch and energy losses showed slightly more variance but stabilized by step 10,000.
- **Overfitting / Instability:** No significant overfitting was observed; the validation loss closely tracked the training loss. Slight high-frequency noise in the validation mel-loss was noted, standard for non-autoregressive models predicting complex spectrograms.

4. Observations on Generated Speech

- **Overall Quality:** At 12,000 steps, both models successfully output intelligible speech. The use of Griffin-Lim introduces a characteristic metallic/robotic artifact compared to neural vocoders, but the phonetic pronunciation is remarkably accurate.
- **English Test Sentences:**
    - **E1 & E2a (Noun/Verb Stress):** The model correctly differentiated the stress pattern in "conduct" (CON-duct as a noun vs. con-DUCT as a verb) based on the supplied phonemes.
    - **E2b (10:00 AM):** Processed as "ten zero zero A M", the synthesis was clear and pacing was accurate.
    - **E4 (Antidisestablishmentarianism):** The model successfully traversed the long, complex multi-syllabic word without duration breakdown or mumbling, proving the robustness of the phoneme-to-mel alignment.

5. Variance Adaptor Plots Analysis

- **Duration:** The predicted duration arrays match the target phoneme sequences flawlessly. Longer duration bars are visibly correctly aligned with vowels and diphthongs, while plosives (e.g., /t/, /p/) show sharp, short durations.
- **Pitch (F0):**
    - In **E1**, the pitch naturally falls at the end of the declarative sentence.
    - In **E3**, pitch variation is notably higher around the word "unexpectedly", mimicking natural human prosody.
- **Energy:** Energy plots exhibit distinct peaks during voiced phonemes (vowels) and drop to near-zero during pauses and voiceless fricatives. This correlation is particularly visible in the stop gaps of E4's lengthy phoneme string.

6. Comparative Observations (English vs. Telugu)

- **Linguistic Differences:** Telugu, being a highly agglutinative and phonetically transparent language, yielded slightly smoother duration alignments out-of-the-box compared to English, which suffers from complex phonotactics and silent letters.
- **Limitations:** The primary limitation for both languages is the Griffin-Lim vocoder, which caps the naturalness (MOS score) of the audio. Furthermore, at 12,000 steps, English prosody is still slightly flat compared to a fully converged model (typically 100k+ steps).
- **Improvements:** Integrating HiFi-GAN or MelGAN and training for up to 200,000 steps would dramatically reduce the robotic artifacts and improve the spectral richness of the voice.

7. Reproducibility

- **Dependencies:** Python 3.8+, PyTorch 2.6, PyWorld, Librosa, unidecode, pypinyin, matplotlib, scipy.
- **Commands:**
  - Preprocessing: `python preprocess.py -p config/<dataset>/preprocess.yaml`
  - Training: `python train.py -p config/<dataset>/preprocess.yaml -m config/<dataset>/model.yaml -t config/<dataset>/train.yaml`
  - Inference: `python inference.py --restore_step 12000` (outputs test wavs and plots based on raw phoneme inputs).